監控與日誌

Monitoring & Logging

AWS Certified DevOps Engineer – Professional (DOP-C02) ｜ 官方考試權重 15%

第 1 題 （題庫 No.27 · 複選（Select 3））

A company is storing 100 GB of log data in csv format in an Amazon S3 bucket SQL developers want to query this data and generate graphs to visualize it. The SQL developers also need an efficient automated way to store metadata from the csv file. Which combination of steps will meet these requirements with the LEAST amount of effort? (Select THREE.)

A. Fitter the data through AWS X-Ray to visualize the data.

B. Filter the data through Amazon QuickSight to visualize the data.

C. Query the data with Amazon Athena.

D. Query the data with Amazon Redshift.

E. Use the AWS Glue Data Catalog as the persistent metadata store.

F. Use Amazon DynamoDB as the persistent metadata store.

我的答案：_____

第 2 題 （題庫 No.52 · 單選）

A highly regulated company has a policy that DevOps engineers should not log in to their Amazon EC2 instances except in emergencies. It a DevOps engineer does log in the security team must be notified within 15 minutes of the occurrence. Which solution will meet these requirements ' ?

A. Install the Amazon Inspector agent on each EC2 instance Subscribe to Amazon EventBridge notifications Invoke an AWS Lambda function to check if a message is about user logins If it is send a notification to the security team using Amazon SNS.

B. Install the Amazon CloudWatch agent on each EC2 instance Configure the agent to push all logs to Amazon CloudWatch Logs and set up a CloudWatch metric filter that searches for user logins. If a login is found send a notification to the security team using Amazon SNS.

C. Set up AWS CloudTrail with Amazon CloudWatch Logs. Subscribe CloudWatch Logs to Amazon Kinesis Attach AWS Lambda to Kinesis to parse and determine if a log contains a user login If it does, send a notification to the security team using Amazon SNS.

D. Set up a script on each Amazon EC2 instance to push all logs to Amazon S3 Set up an S3 event to invoke an AWS Lambda function which invokes an Amazon Athena query to run. The Athena query checks tor logins and sends the output to the security team using Amazon SNS.

我的答案：_____

第 3 題 （題庫 No.89 · 單選）

A company is migrating an application to Amazon Elastic Container Service (Amazon ECS). The company wants to consolidate log data in Amazon CloudWatch in the us-west-2 Region. No CloudWatch log groups currently exist for Amazon ECS. The company receives the following error code when an ECS task attempts to launch: "service my-service-name was unable to place a task because no container instance met all of its requirements." The ECS task definition includes the following container log configuration: " logConfiguration " : { " logDriver " : " awslogs " , " options " : { " awslogs-create-group " : " true " , " awslogs-group " : " awslogs-mytask " , " awslogs-region " : " us-west-2 " , " awslogs-stream-prefix " : " awslogs-mytask " , " mode " : " non-blocking " , " max-buffer-size " : " 25m " } } The ECS cluster uses an Amazon EC2 Auto Scaling group to provide capacity for tasks. EC2 instances launch an Amazon ECS-optimized AMI. Which solution will fix the problem?

A. Modify the ECS infrastructure IAM role to add the logs:CreateLogStream and logs:PutLogEvents permissions.

B. Modify the ECS log configuration to use blocking mode.

C. Modify the ECS container instance IAM role to add the logs:CreateLogStream and logs:PutLogEvents permissions.

D. Modify the ECS log configuration by setting the awslogs-create-group option to false.

我的答案：_____

第 4 題 （題庫 No.112 · 複選（Select 2））

A DevOps engineer notices that all Amazon EC2 instances running behind an Application Load Balancer in an Auto Scaling group are failing to respond to user requests. The EC2 instances are also failing target group HTTP health checks Upon inspection, the engineer notices the application process was not running in any EC2 instances. There are a significant number of out of memory messages in the system logs. The engineer needs to improve the resilience of the application to cope with a potential application memory leak. Monitoring and notifications should be enabled to alert when there is an issue Which combination of actions will meet these requirements? (Select TWO.)

A. Change the Auto Scaling configuration to replace the instances when they fail the load balancer ' s health checks.

B. Change the target group health check HealthCheckIntervalSeconds parameter to reduce the interval between health checks.

C. Change the target group health checks from HTTP to TCP to check if the port where the application is listening is reachable.

D. Enable the available memory consumption metric within the Amazon CloudWatch dashboard for the entire Auto Scaling group Create an alarm when the memory utilization is high Associate an Amazon SNS topic to the alarm to receive notifications when the alarm goes off

E. Use the Amazon CloudWatch agent to collect the memory utilization of the EC2 instances in the Auto Scaling group Create an alarm when the memory utilization is high and associate an Amazon SNS topic to receive a notification.

我的答案：_____

第 5 題 （題庫 No.148 · 單選）

A DevOps engineer manages a large commercial website that runs on Amazon EC2. The website uses Amazon Kinesis Data Streams to collect and process web togs. The DevOps engineer manages the Kinesis consumer application, which also runs on Amazon EC2. Sudden increases of data cause the Kinesis consumer application to (all behind and the Kinesis data streams drop records before the records can be processed. The DevOps engineer must implement a solution to improve stream handling. Which solution meets these requirements with the MOST operational efficiency?

A. Modify the Kinesis consumer application to store the logs durably in Amazon S3 Use Amazon EMR to process the data directly on Amazon S3 to derive customer insights Store the results in Amazon S3.

B. Horizontally scale the Kinesis consumer application by adding more EC2 instances based on the Amazon CloudWatch GetRecords IteratorAgeMilliseconds metric Increase the retention period of the Kinesis data streams.

C. Convert the Kinesis consumer application to run as an AWS Lambda function. Configure the Kinesis data streams as the event source for the Lambda function to process the data streams

D. Increase the number of shards in the Kinesis data streams to increase the overall throughput so that the consumer application processes the data faster.

我的答案：_____

第 6 題 （題庫 No.175 · 單選）

A company has many AWS accounts. During AWS account creation the company uses automation to create an Amazon CloudWatch Logs log group in every AWS Region that the company operates in. The automaton configures new resources in the accounts to publish logs to the provisioned log groups in their Region. The company has created a logging account to centralize the logging from all the other accounts. A DevOps engineer needs to aggregate the log groups from all the accounts to an existing

Amazon S3 bucket in the logging account. Which solution will meet these requirements in the MOST operationally efficient manner?

A. In the logging account create a CloudWatch Logs destination with a destination policy. For each new account subscribe the CloudWatch Logs log groups to the. Destination Configure a single Amazon Kinesis data stream and a single Amazon Kinesis Data Firehose delivery stream to deliver the logs from the CloudWatch Logs destination to the S3 bucket.

B. In the logging account create a CloudWatch Logs destination with a destination policy for each Region. For each new account subscribe the CloudWatch Logs log groups to the destination. Configure a single Amazon Kinesis data stream and a single Amazon Kinesis Data Firehose delivery stream to deliver the logs from all the CloudWatch Logs destinations to the S3 bucket.

C. In the logging account create a CloudWatch Logs destination with a destination policy for each Region. For each new account subscribe the CloudWatch Logs log groups to the destination Configure an Amazon Kinesis data stream and an Amazon Kinesis Data Firehose delivery stream for each Region to deliver the logs from the CloudWatch Logs destinations to the S3 bucket.

D. In the logging account create a CloudWatch Logs destination with a destination policy. For each new account subscribe the CloudWatch Logs log groups to the destination. Configure a single Amazon Kinesis data stream to deliver the logs from the CloudWatch Logs destination to the S3 bucket.

我的答案：_____

第 7 題 （題庫 No.200 · 複選（Select 2））

A DevOps engineer is implementing governance controls for a company that requires its infrastructure to be housed within the United States. The engineer must restrict which AWS Regions can be used, and ensure an alert is sent as soon as possible if any activity outside the governance policy takes place. The controls should be automatically enabled on any new Region outside the United States (US). Which combination of actions will meet these requirements? (Select TWO.)

A. Create an AWS Organizations SCP that denies access to all non-global services in non-US Regions. Attach the policy to the root of the organization.

B. Configure AWS CloudTrail to send logs to Amazon CloudWatch Logs and enable it for all Regions. Use a CloudWatch Logs metric filter to send an alert on any service activity in non-US Regions.

C. Use an AWS Lambda function that checks for AWS service activity and deploy it to all Regions. Write an Amazon EventBridge rule that runs the Lambda function every hour, sending an alert if activity is found in a non-US Region.

D. Use an AWS Lambda function to query Amazon Inspector to look for service activity in non-US Regions and send alerts if any activity is found.

E. Write an SCP using the aws: RequestedRegion condition key limiting access to US Regions. Apply the policy to all users, groups, and roles

我的答案：_____

第 8 題 （題庫 No.246 · 單選）

A company uses machine learning on transactional data that a development team uploads from multiple sources to an Amazon S3 bucket. The development team finds duplicate objects in the S3 bucket across multiple prefixes. The duplications are increasing the company ' s storage costs. The company wants to implement an automated solution to identify and remove duplicate objects from the S3 bucket. Which solution will meet this requirement?

A. Use Amazon S3 Storage Lens to scan the S3 bucket and prefixes, create an inventory of the duplicate objects, and invoke an AWS Lambda function to delete the duplicate objects.

B. Configure AWS CloudTrail to record data events for the S3 bucket. Use Amazon Athena to query S3 data events to identify duplicate objects. Use an AWS Lambda function to delete duplicate objects.

C. Use Amazon S3 Inventory to generate reports. Use Amazon Athena to query the reports based on ETag values. Use S3 Batch Operations and an AWS Lambda function to remove duplicate objects.

D. Use AWS Config to identify object changes. Use Amazon EventBridge to delete duplicate objects by running the aws:executeScript action in AWS Systems Manager.

我的答案：_____

第 9 題 （題庫 No.293 · 單選）

A company runs a microservices application on Amazon EKS. Users report delays accessing an account summary feature during peak hours. CloudWatch metrics and logs show normal CPU and memory utilization on EKS nodes. The DevOps engineer cannot identify where delays occur within the microservices. Which solution will meet these requirements?

A. Deploy the AWS X-Ray daemon as a DaemonSet in the EKS cluster. Use the X-Ray SDK to instrument the application code. Redeploy the application.

B. Enable CloudWatch Container Insights for the EKS cluster. Use the Container Insights data to diagnose delays.

C. Create alarms based on existing CloudWatch metrics. Set up SNS email alerts.

D. Increase the timeout settings in the application code for network operations.

我的答案：_____

第 10 題 （題庫 No.400 · 單選）

A company runs a development environment website and database on an Amazon EC2 instance that uses Amazon Elastic Block Store (Amazon EBS) storage. The company wants to make the instance more resilient to underlying hardware issues. The company wants to automatically recover the EC2 instance if AWS determines the instance has lost network connectivity. Which solution will meet these requirements?

A. Add the EC2 instance to an Auto Scaling group. Set the minimum, maximum, and desired capacity to 1.

B. Add the EC2 instance to an Auto Scaling group. Configure a lifecycle hook to detach the EBS volume if the EC2 instance shuts down or terminates.

C. Create an Amazon CloudWatch alarm for the StatusCheckFailed_System metric. Add an EC2 action to recover the instance when the alarm state is in ALARM.

D. Create an Amazon CloudWatch alarm for the NetworkOut metric. Add an EC2 action to recover the instance when the alarm state is in INSUFFICIENT_DATA.

我的答案：_____

解答與解析　Answer Key

解析取自題庫原文，未經 AWS 官方審核；若與官方文件有出入，以官方文件為準。

第 1 題（No.27）　正確答案：B、C、E

https://docs.aws.amazon.com/glue/latest/dg/components-overview.html

涉及服務：X-Ray、S3、DynamoDB、Athena

第 2 題（No.52）　正確答案：B

https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/

涉及服務：Lambda、EC2/Auto Scaling、CloudWatch、CloudTrail、S3、SNS

第 3 題（No.89）　正確答案：C

When using the awslogs log driver with Amazon ECS on EC2, CloudWatch Logs permissions must be granted to the ECS container instance IAM role , not the task definition or infrastructure role. The ECS agent running on the EC2 instances is responsible for creating log streams and pushing log events to Amazon CloudWatch Logs on behalf of the containers.

In this scenario, the task definition is correctly configured to automatically create the log group (awslogs- create-group: true) and send logs to the specified Region. However, the error occurs because the EC2 container instances do not have sufficient IAM permissions to perform the required CloudWatch Logs API calls. As a result, ECS cannot place the task, and it reports that no container instance meets the requirements.

According to AWS documentation, the container instance IAM role must include the following permissions when using the awslogs driver:

* logs:CreateLogStream
* logs:PutLogEvents
* (and, when creating log groups automatically) logs:CreateLogGroup

Option C correctly addresses the root cause by updating the container instance IAM role . Option A is incorrect because the ECS infrastructure or service role is not used to write logs. Option B is unrelated to permissions and does not resolve the issue. Option D would fail because the log group does not already exist, causing task startup to fail.

Therefore, modifying the container instance IAM role is the correct solution.

涉及服務：EC2/Auto Scaling、ECS、CloudWatch、IAM/SCP

第 4 題（No.112）　正確答案：A、E

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/metrics-collected-by-CloudWatch-agent.

html

涉及服務：EC2/Auto Scaling、CloudWatch、SNS、ELB/ALB/NLB

第 5 題（No.148）　正確答案：B

https://docs.aws.amazon.com/streams/latest/dev/monitoring-with-cloudwatch.html

GetRecords.IteratorAgeMilliseconds - The age of the last record in all GetRecords calls made against a Kinesis stream, measured over the specified time period. Age is the difference between the current time and when the last record of the GetRecords call was written to the stream. The Minimum and Maximum statistics can be used to track the progress of Kinesis consumer applications. A value of zero indicates that the records being read are completely caught up.

涉及服務：Lambda、EC2/Auto Scaling、CloudWatch、S3、Kinesis

第 6 題（No.175） 正確答案：C

This solution will meet the requirements in the most operationally efficient manner because it will use CloudWatch Logs destination to aggregate the log groups from all the accounts to a single S3 bucket in the logging account. However, unlike option A, this solution will create a CloudWatch Logs destination for each region, instead of a single destination for all regions. This will improve the performance and reliability of the log delivery, as it will avoid cross-region data transfer and latency issues. Moreover, this solution will use an Amazon Kinesis data stream and an Amazon Kinesis Data Firehose delivery stream for each region, instead of a single stream for all regions. This will also improve the scalability and throughput of the log delivery, as it will avoid bottlenecks and throttling issues that may occur with a single stream.

涉及服務：CloudWatch、S3、Kinesis

第 7 題（No.200） 正確答案：A、B

To implement governance controls that restrict AWS service usage to within the United States and ensure alerts for any activity outside the governance policy, the following actions will meet the requirements:

A). Create an AWS Organizations SCP that denies access to all non-global services in non-US Regions. Attach the policy to the root of the organization. This action will effectively prevent users and roles in all accounts within the organization from accessing services in non-US Regions12.

B). Configure AWS CloudTrail to send logs to Amazon CloudWatch Logs and enable it for all Regions. Use a CloudWatch Logs metric filter to send an alert on any service activity in non-US Regions. This action will allow monitoring of all AWS Regions and will trigger alerts if any activity is detected in non-US Regions, ensuring that the governance team is notified as soon as possible3.

AWS Documentation on Service Control Policies (SCPs) and how they can be used to manage permissions and restrict access based on Regions12.

AWS Documentation on monitoring CloudTrail log files with Amazon CloudWatch Logs to set up alerts for specific activities3.

涉及服務：Lambda、CloudWatch、CloudTrail、Organizations/Control Tower、IAM/SCP、EventBridge

第 8 題（No.246） 正確答案：C

Amazon S3 Inventory generates scheduled reports listing all objects in a bucket with metadata including the ETag, which is an MD5 hash of the object content. Objects with the same ETag value have identical content regardless of their key or prefix location and can therefore be identified as duplicates. Amazon Athena can query the inventory report stored in S3 using SQL to group objects by ETag and identify keys where more than one object shares the same hash. Once duplicate keys are identified, S3 Batch Operations can invoke an AWS Lambda function to programmatically delete the duplicate entries, completing the automated cleanup workflow. S3 Storage Lens (Option A) provides visibility into storage usage metrics but does not identify content-level duplicates. CloudTrail (Option B) tracks API activity but cannot compare object content. AWS Config (Option D) is designed for resource configuration compliance, not S3 object deduplication.

涉及服務：Systems Manager (SSM)、Lambda、CloudTrail、Config、S3、EventBridge

第 9 題（No.293） 正確答案：A

* AWS X-Ray provides distributed tracing, which allows visualization of latencies and errors within microservices, pinpointing bottlenecks or delays.

* Instrumenting application code with the X-Ray SDK and running the X-Ray daemon as a DaemonSet in EKS ensures tracing data is collected cluster-wide.

* Container Insights (Option B) provides resource-level metrics but not detailed request tracing.

* CloudWatch alarms and alerts (Option C) detect symptoms but don ' t provide root cause tracing.

* Increasing timeouts (Option D) only masks the issue and does not diagnose it.

References:

AWS X-Ray for Amazon EKS
Distributed Tracing in Microservices
涉及服務：EKS、CloudWatch、X-Ray、SNS

第 10 題（No.400） 正確答案：C

Configure a CloudWatch alarm on StatusCheckFailed_System and set the EC2 recovery action. Instance recovery automatically recovers the instance on impaired hardware while preserving the instance ID, private IPs, EBS volumes, and metadata.
涉及服務：EC2/Auto Scaling、CloudWatch